1. Clinical Question & Core Hypothesis

Clinical Question

Why do only some MIBC tumors respond to intravesical BCG despite similar clinical staging—specifically, does the organization and density of peritumoral lymphocytes at the invasive margin (visible on H&E) determine whether BCG can elicit an effective local anti-tumor immune response? 2) Tier 1 path built on: the single provided path (Node 1 → Node 2 → Node 3), i.e., that the PIM-Score (an H&E-derived summary of peritumoral lymphocyte density and spatial organization) captures the local immune architecture that mechanistically enables BCG-mediated tumor control. I built on this because the PIM-Score directly operationalizes the hypothesized mediator (margin-local immune readiness) between baseline histology and BCG outcome. 3) Tier 2/supporting evidence: the PIM feature components themselves (radial density peak near margin, clustering metrics, and margin irregularity) provide mechanistic support by distinguishing mere presence of lymphocytes from an organized, margin-focused immune architecture likely required for effective antigen presentation and effector function; the optional TSR-at-margin supports the microenvironmental context (stroma-rich vs tumor-dense) that can modulate lymphocyte access/function. 4) Endpoint choice: I selected therapy_response because the clinical question is explicitly about predicting BCG efficacy (responder vs non-responder or recurrence-free survival after BCG), making therapy_response the most directly relevant and actionable endpoint. 5) Deliberate exclusions: I did not include any molecular genes because none were provided by the ontologist and the user specified an H&E-only biomarker; I excluded broader endpoints (OS, PFI) to keep the question focused on BCG responsiveness; I also excluded TLS-like aggregates and additional imaging biomarkers from the core hypothesis to obey the constraint of exactly one imaging feature in the mechanistic claim, while preserving them (TSR) as an optional secondary modifier?

Working Hypothesis

Pre-treatment PIM-score, derived exclusively from H&E–based assessment of peritumoral TIL density and spatial organization at the invasive margin, predicts therapy_response to intravesical BCG in bladder cancer. Specifically, tumors with a high PIM-score—characterized by a dense radial TIL density peak within 0–200 μm of the margin, organized margin-focused TIL clustering, and inflamed-margin morphology—will show higher odds of favorable therapy_response to BCG, independent of standard clinical covariates. This predictive value is modulated by the stromal context at the margin, as captured by tumor-stroma ratio (TSR) at the invasive margin, which may enhance discrimination between responders and non-responders. Anchor_genes remain unused due to ontologist constraints, and TLS-like features are not required for the primary claim but can be explored as secondary modifiers in future studies.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Tumor-stroma ratio (TSR) at the invasive margin	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: assemble two independent retrospective cohorts with pre-treatment H&E and documented BCG outcomes;

Prediction 2: compute PIM-score from digital pathology by quantifying (i) radial TIL density within 0–200 μm of the invasive margin, (ii) margin-focused TIL clustering (e.g., Moran's I or Ripley's K near the margin), and (iii) margin irregularity;

Prediction 3: model therapy_response using logistic regression or survival methods, adjusting for clinical covariates, with TSR as a modifier;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Tumor-stroma ratio (TSR) at the invasive margin: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.